

SUMMARY OF SHORTING JUMPERS AND BLINKER LED2

SJ2 for PS	SJ1 for OPT	Mode	Operation	Blinker (LED2)
High	High	TX	Continuous data transmission	Blinks low twice
High	Low	TX	Manual triggering for data transmission	Blinks low once
Low	High	RX	Decoding enabled (4-bit code+8-bit data)	Blinks high twice
Low	Low	RX	Decoding disabled (12-bit data)	Blinks high once

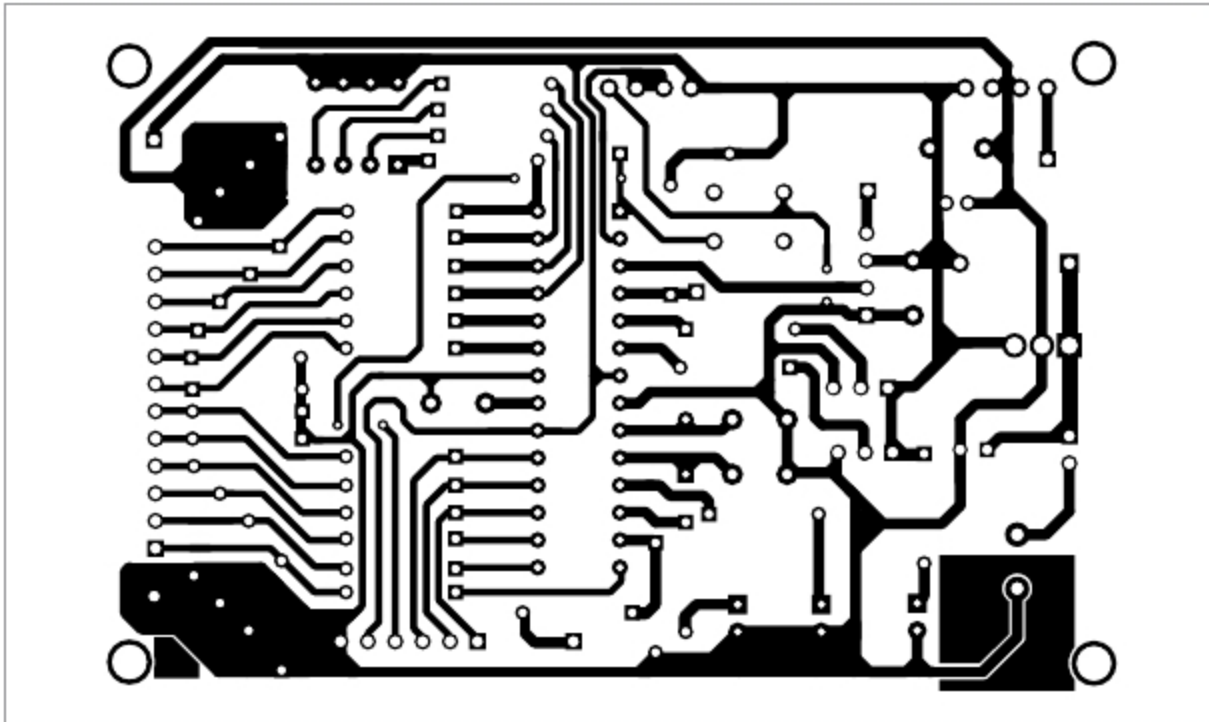


Fig. 2: Actual-size PCB layout of 12-bit signal transmitter and receiver

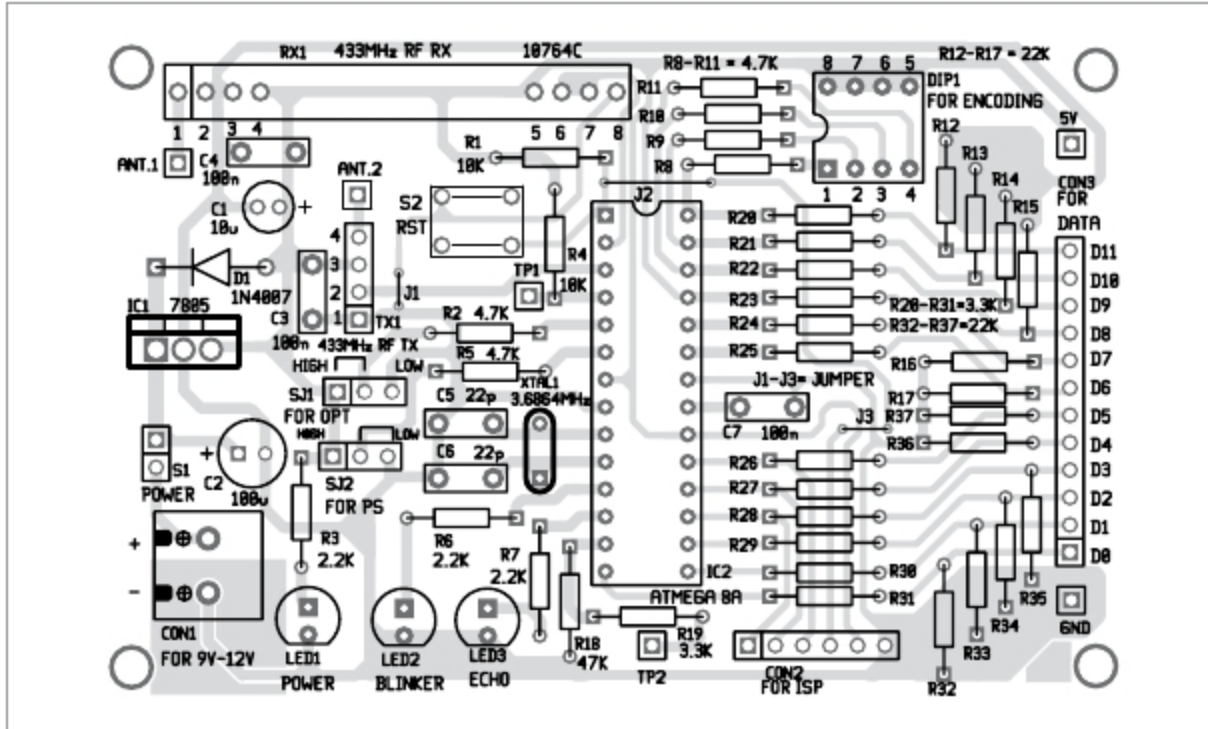


Fig. 3: Components layout for the PCB of 12-bit signal transmitter and receiver

program code (`#define LOOP_DELAY 2000//milli seconds`). If SJ1 is low, transmission is activated with falling edge on pin 4 (PD2) of MCU.

In RX mode, if SJ1 is high, decoding is active. And if SJ1 is low, decoding is inactive. Once decoding is active, the four MSB (D8 to D11) or address codes (A0 to A3) of receiver side should match the four MSB (D8

to D11) or address codes (A0 to A3) of transmitter side. Output data is available on D0 to D7 (8 bits only) of CON3.

Receiver side does not respond to the transmission signal if address codes or MSBs are mismatched. Four MSB codes and eight LSB data (total 12-bit data) are transmitted, and 16 such data sets can be transmitted from

the transmitter unit without interference.

Blinker LED2 blinks once in RX mode without decoding and twice with it. It goes low once in TX mode with manual triggering and twice with continuous data transmission.

Echo LED3 blinks whenever data is transmitted or received. Output available at TP1 can be used for enabling/latching the new data available on D0-D11 pins, if required.

Manual triggering is low and can be applied at TP1. On receiving new data by receiver circuit, TP2 is active high for about 500 microseconds.

To understand decoding on receiver side, set SJ1 to high and connect a 4-way DIP switch. Then, set 4-bit (A0-A3) address on the DIP switch. Now, send data from TX side. RX side receives 8-bit data, if A0-A3 settings for both transmitter and receiver match; otherwise, 8-bit data is not received.

Software

The program (`parallelsignal.c`) is written in C and compiled using AVR Studio4. After the circuit connections are done, write/burn `parallelsignal.hex` file to ATmega8A MCU using a suitable AVR programmer through ISP port CON2.

There is no fuse-bit setting in the programmer. By default, low fuse bit is E1 and high fuse bit is D9. Fuse-bit settings for crystal are required if 3.6864MHz (XTAL1) crystal is connected, which is optional. Use the same crystal in both transmitter and receiver boards to achieve synchronisation of serial data transfer.

Construction and testing

An actual-size PCB layout for the 12-bit signal transmitter and receiver is shown in Fig. 2 and its components layout in Fig. 3. Assemble two sets (boards) of the designed PCB: one for transmitter board with TX1 and another for receiver board with RX1. Note that, RX1 is not required in transmitter board and TX1 is not required in receiver board.

A separate DC power supply (9V-12V) is required for transmitter and receiver boards. Upload `ParallelSerial.hex` code to the respective MCUs of